

Disproof of conjecture 00000002175

earthking11

2026-09-13

Verdict: FALSE

We disprove conjecture 00000002175 as stated. The refutation is unconditional and elementary, and is already complete on a four-element ground set: there is a 3-wise odd-intersecting family of size 4, whereas the conjecture asserts the maximum is $2^{n-3} = 2$. Computations of the exact maxima for $n \leq 7$ contradict the claimed formula in both directions, and an exhaustive check of all affine cosets shows that no affine coset is an extremal family, so the characterisation clause is false as well.

1 The conjecture

Definition 1 (3-wise odd-intersecting). A family $\mathcal{F} \subseteq \mathcal{P}([n])$ of subsets of an n -element ground set $[n] = \{1, \dots, n\}$ is *3-wise odd-intersecting* if for every three *distinct* members $A, B, C \in \mathcal{F}$ the intersection $A \cap B \cap C$ has odd cardinality.

Quoted from `conjectures/00000002175.md`:

Definition: A 3-wise odd-intersecting family is one where the intersection of any three members is odd. **Conjecture:** The maximal size is always 2^{n-3} ; and the extremal families consist of indicator functions of 3-dimensional affine subspaces. (3-wise odd-intersection affine extremal)

The original statement is filed in English and Chinese; the Chinese text reads (the conjecture-file version is reproduced verbatim in `README.md`). The size claim (the maximum is always 2^{n-3}) and the characterisation claim (the extremal families are the indicator functions of 3-dimensional affine subspaces) both fail.

2 The counterexample at $n = 4$

Theorem 2. *On the ground set $\{1, 2, 3, 4\}$, the four-member family*

$$F = \{ \{1, 2\}, \{1, 3\}, \{1, 4\}, \{1, 2, 3, 4\} \}$$

is 3-wise odd-intersecting, yet $|F| = 4 > 2 = 2^{4-3}$.

Proof. There are $\binom{4}{3} = 4$ triples of distinct members, and each meets in exactly $\{1\}$, of odd size 1:

$$\begin{aligned} \{1, 2\} \cap \{1, 3\} \cap \{1, 4\} &= \{1\}, & \{1, 2\} \cap \{1, 3\} \cap \{1, 2, 3, 4\} &= \{1\}, \\ \{1, 2\} \cap \{1, 4\} \cap \{1, 2, 3, 4\} &= \{1\}, & \{1, 3\} \cap \{1, 4\} \cap \{1, 2, 3, 4\} &= \{1\}. \end{aligned}$$

Indeed, every one of the four members contains 1 (so every intersection contains 1), and no other point lies in all three members of any triple. Hence F is 3-wise odd-intersecting. Since $|F| = 4$ while $2^{4-3} = 2$, the size claim of the conjecture fails. $\square$

Corollary 3. *The true maximum at $n = 4$ is at least 5, so the conjecture's value $2^{4-3} = 2$ is too small by at least 3. A 5-element witness is*

$$\{ \{1\}, \{1, 2\}, \{1, 3\}, \{1, 4\}, \{1, 2, 3\} \},$$

in which every triple of distinct members meets in $\{1\}$ (a triple containing $\{1\}$ always meets in $\{1\}$; among the remaining four sets, which all contain 1, any three have intersection exactly $\{1\}$).

The conjecture is thus refuted by the four-member family alone; the five-member family shows the failure is even larger than the witness indicates.

3 The true maxima versus 2^{n-3}

The exact maximum $M(n)$ was computed in two independent ways (both implemented in `reproduce.py`):

- *Exhaustive search* over all 2^{2^n} subfamilies of $\mathcal{P}([n])$ for $n \leq 4$, with a cardinality prune.
- *Branch and bound* for the maximum independent set of the 3-uniform hypergraph on the 2^n subsets whose hyperedges are the triples with *even* intersection, for $n = 5, 6, 7$. The bound greedily packs vertex-disjoint forbidden triples (each permits at most 2 of its three vertices); vertices are ordered by conflict degree and the conflict structure is precomputed from a parity table, so the $n = 7$ search runs in about ten seconds.

n	true maximum $M(n)$	claim 2^{n-3}	comparison
0	1	— (undefined)	—
1	2	— (undefined)	—
2	2	— (undefined)	—
3	4	1	claim too small by 3
4	5	2	claim too small by 3
5	7	4	claim too small by 3
6	8	8	equal (coincidence only)
7	10	16	claim too large by 6

Table 1: True maxima $M(n)$ for $n = 0, \dots, 7$ versus the claim 2^{n-3} .

Equivalently,

$$M(n) = 1, 2, 2, 4, 5, 7, 8, 10 \quad (n = 0, 1, \dots, 7).$$

The claim is too small at $n = 3, 4, 5$, too large at $n = 7$ (there is no family of size 16; the maximum is 10), and agrees at $n = 6$ only by coincidence. Hence 2^{n-3} is not the truth.

4 The construction

The extremal families for $n = 3, \dots, 7$ have the following shape. Fix a point x and let $Y = [n] \setminus \{x\}$. Put

$$G = \{\emptyset\} \cup \{\{y\} : y \in Y\} \cup \{\text{the edges of a matching on } Y\}, \quad F = \{\{x\} \cup g : g \in G\}.$$

Theorem 4. *The family F above is 3-wise odd-intersecting and has*

$$|F| = 1 + (n-1) + \left\lfloor \frac{n-1}{2} \right\rfloor = n + \left\lfloor \frac{n-1}{2} \right\rfloor.$$

For $n = 3, 4, 5, 6, 7$ this equals 4, 5, 7, 8, 10, i.e. $M(n)$.

Proof. Any three distinct members have the form $\{x\} \cup g_1, \{x\} \cup g_2, \{x\} \cup g_3$ with distinct $g_i \in G$, and

$$(\{x\} \cup g_1) \cap (\{x\} \cup g_2) \cap (\{x\} \cup g_3) = \{x\} \cup (g_1 \cap g_2 \cap g_3).$$

We claim $g_1 \cap g_2 \cap g_3 = \emptyset$. If some $g_i = \emptyset$ this is clear. Otherwise each g_i is a singleton $\{y\}$ or a matching edge. A singleton $\{y\}$ equals at most one other g_j , and at most one matching edge contains y ; since g_1, g_2, g_3 are pairwise distinct, no point y can belong to all three. Two distinct matching edges are disjoint by definition of a matching. Hence the intersection is empty and every triple meets in $\{x\}$, of odd size 1. The cardinality is $1 + |Y| + \lfloor |Y|/2 \rfloor$ with $|Y| = n-1$. $\square$

Thus the extremal families have a “fixed point plus an even-intersecting family on the remainder” shape, not the affine-subspace shape the conjecture proposes.

Remark 5 (Asymptotic lower bound). Partitioning Y into pairs and taking G to be *all* unions of these pairs, the intersection $g_1 \cap g_2 \cap g_3$ is again a union of pairs (hence of even size), so every triple still meets in $\{x\} \cup$ (a union of pairs), of odd size, and

$$|F| = 2^{\lfloor (n-1)/2 \rfloor} \approx 2^{n/2}.$$

This is an infinite family, so $M(n) \geq 2^{\lfloor (n-1)/2 \rfloor}$: the maximum grows at least like $2^{n/2}$, an exponential of rate $n/2$ rather than $n-3$. (The computed values $M(6) = 8 = 2^3$ and $M(7) = 10$ are consistent with growth of that order. The lower bound alone does not determine $M(n)$ for large n ; what is decisive for the conjecture is the table of Section 4, which refutes 2^{n-3} in both directions.)

5 The affine-subspace characterisation is false

The phrase “indicator functions of 3-dimensional affine subspaces” admits two readings. Identify a subset of $[n]$ with its indicator vector in $\mathbb{F}_2^n$, so that a family of indicator vectors is a set of points of $\mathbb{F}_2^n$. Both readings were tested by enumerating *all* affine cosets of *all* dimensions in $\mathbb{F}_2^n$ for $n = 4, 5, 6, 7$.

Literal reading: a 3-dimensional affine subspace. Such a coset has $2^3 = 8$ members. Enumerating all 3-dimensional subspaces and all their cosets:

n	3-dim. subspaces	cosets	cosets that are 3-wise odd
4	15	30	0
5	155	620	0
6	1395	11160	0
7	11811	188976	105

For $n = 4, 5, 6$ there is *no* 3-dimensional affine coset that is 3-wise odd-intersecting. For $n = 7$ there are 105 such cosets, but every one has size $8 < M(7) = 10$, so none is extremal.

Codimension-3 reading: a coset of size 2^{n-3} . Here $n = 4$: cosets of size 2, all vacuously 3-wise odd, but $2 < M(4) = 5$; $n = 5$: cosets of size 4, of which 460 are 3-wise odd, but $4 < M(5) = 7$; $n = 6$: cosets of size 8, and *none* is 3-wise odd; $n = 7$: such a coset would have size $16 > M(7) = 10$, impossible.

Corollary 6. *No affine coset of any dimension is an extremal family for $n = 4, 5, 6, 7$. If $n \in \{4, 5, 7\}$ then $M(n) = 5, 7, 10$ is not a power of two, while every affine coset has power-of-two size, so no coset can even have $M(n)$ members. At $n = 6$, where $M(6) = 8 = 2^3$ is a power of two, the only cosets of size 8 are 3-dimensional, and none of them is 3-wise odd-intersecting.*

Remark 7 (Ambiguity note). “Indicator functions of 3-dimensional affine subspaces” is garbled: a 3-dimensional affine subspace has $2^3 = 8$ points, whereas 2^{n-3} is the size of a codimension-3 coset. Each indicator function of a 3-dimensional affine subspace is a *member* of the proposed extremal family, while 2^{n-3} is a *number of members*; the two readings are different statements, and neither yields valid extremal families.

Remark 8. The $n = 4$ witness is itself an affine coset, but of dimension 2: its indicator vectors satisfy $F = \{1100, 1010, 1001, 1111\} = 1100 + \{0000, 0110, 0101, 0011\}$, a coset of the span of 0110 and 0101. Its size is $2^2 = 4$, its dimension is 2 (not 3), and it is not extremal because $M(4) = 5$. So being an affine coset is neither sufficient nor, at the claimed dimension, possible.

6 Formalisation

The $n = 4$ witness is formalised in the accompanying Lean 4 project under `lean4/`, using core Lean only (`import Std`, no `Mathlib`, no `sorry`). Members are represented as indicator vectors of type `List Bool`. The main statements are

```
def interSize (a b c : List Bool) : Nat
def TripleOdd (F : List (List Bool)) : Prop
def F : List (List Bool) :=
  [[true,true,false,false], [true,false,true,false],
   [true,false,false,true], [true,true,true,true]]
theorem witness_oddTriples : TripleOdd F
theorem triple_123 : interSize m12 m13 m14 = 1
theorem triple_124 : interSize m12 m13 m1234 = 1
theorem triple_134 : interSize m12 m14 m1234 = 1
theorem triple_234 : interSize m13 m14 m1234 = 1
theorem witness_length : F.length = 4
theorem witness_breaks_bound : F.length > 2 ^ (4 - 3 : Nat)
theorem witness_check : checkFamily F = true
theorem n4_max_at_least_five :
  TripleOdd G5 ∧ G5.length = 5 ∧ G5.length > 2 ^ (4 - 3 : Nat)
```

Here `TripleOdd` is the conjecture’s property, `F` is the four-member witness, and `G5` is the five-member family of Corollary 3.1. The four triple-intersection theorems are the explicit check that the witness meets the property; `witness_breaks_bound` is the inequality $4 > 2$. Building and auditing:

```
$ cd lean4
$ lake build
```

```
$ lake env lean Check.lean
```

The audit reports no `sorryAx`; the concrete computational theorems depend on no axioms at all, and the quantified ones use only `propext` and `Quot.sound`. The exact maxima $M(n)$ and the affine-coset enumeration are finite computations carried out by `reproduce.py`; they are not needed for the refutation, which is already complete at $n = 4$. See `lean4/README.md` for the full statement table and scope.

7 Reproducibility

The checks are carried out by `reproduce.py`, which uses the Python 3 standard library only:

```
$ python3 reproduce.py
... (the witness, the maxima, the constructions, the affine cosets)
PASS: all checks verified; conjecture 00000002175 is FALSE.
```

It verifies the witness's four triple intersections (each equal to $\{1\}$), computes $M(0..4)$ exhaustively and $M(5..7)$ by branch and bound (cross-checking the two methods at $n = 4$), checks the constructions of Section 5, and tests every affine coset for $n = 4, \dots, 7$. It prints `PASS` and exits 0 exactly when every check holds. The document itself is built with `tectonic --outdir build main.tex`.